Multi-band GNSS / NavIC Active Antenna - RF Front End

Technical note: architecture, simulated RF performance, interference risk and actions (L1, L2/L5, NavIC S)

Doc. No.: _____ | Rev.: Draft A | Date: 2026-10-31

Field	Entry
Document	Technical note - RF front end of the multi-band active antenna
Document No. / Rev.	_____ / Draft A
Prepared / Checked / Approved	_____ / _____ / _____
Classification	BEL internal. Technical details for internal circulation only.
Basis	Manufacturer data sheets of the selected parts (listed in section 9); circuit simulation. No hardware measurement yet.

Document control.

1 Scope

This note gives the RF design data of the active antenna front end: the antenna element is only mentioned (dual-feed RHCP patch per band, modelled as an ideal source); everything behind the feeds - hybrid combiners, LNAs, SAW filters and the band combiner - is modelled from the manufacturer data and simulated. Three bands: GNSS L1 (B1I/L1/E1/G1) and L2/L5 (L5/E5a/E5b/L2) on one combined output, and NavIC S-band (2492.028 MHz) on a separate output. All LNAs at +5 V.

2 Architecture

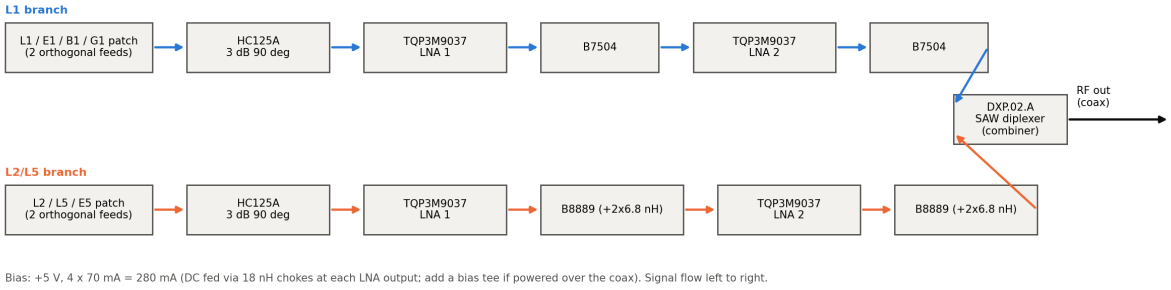


Figure 1 - L-band chain (two branches combined in the DXP.02.A).

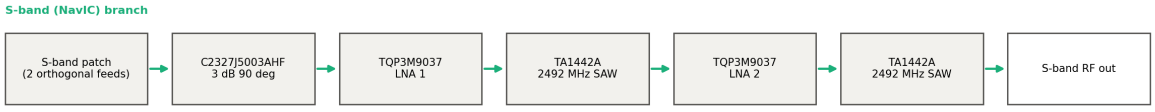


Figure 2 - NavIC S-band chain (separate output).

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

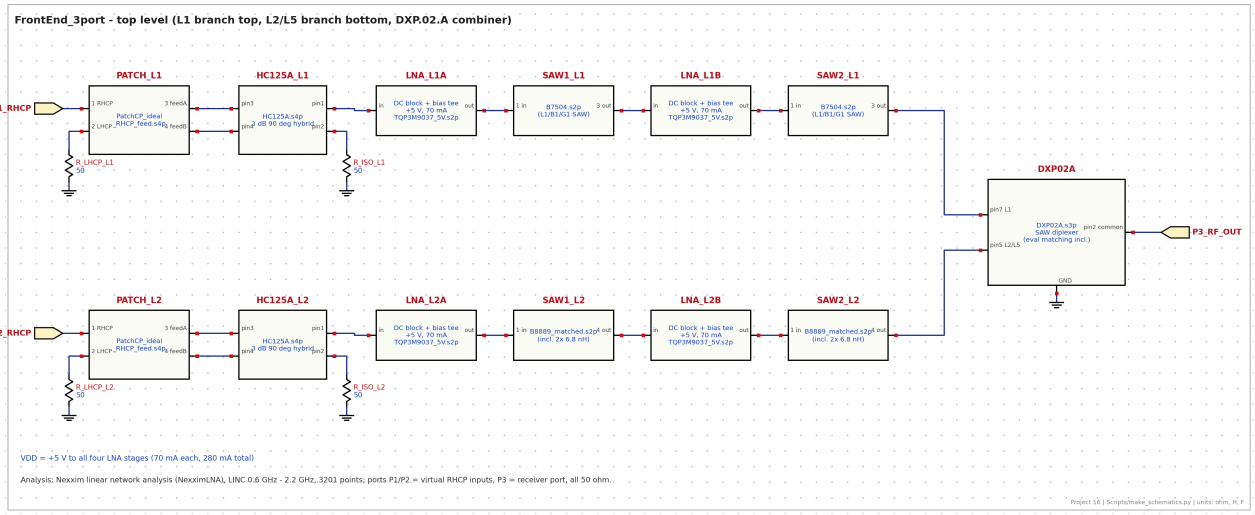


Figure 3 - Simulation schematic, L-band (design FrontEnd_3port).

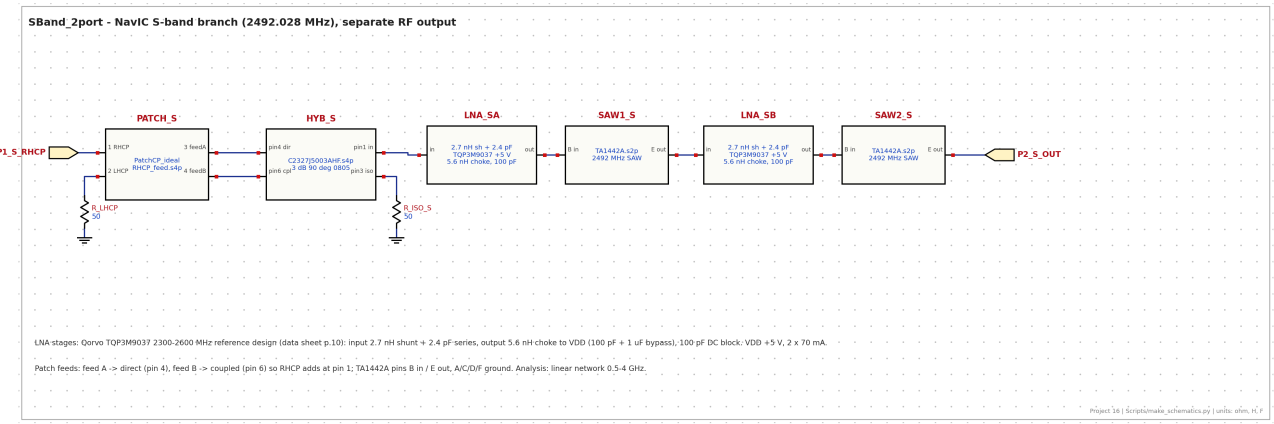


Figure 4 - Simulation schematic, S-band (design SBand_2port).

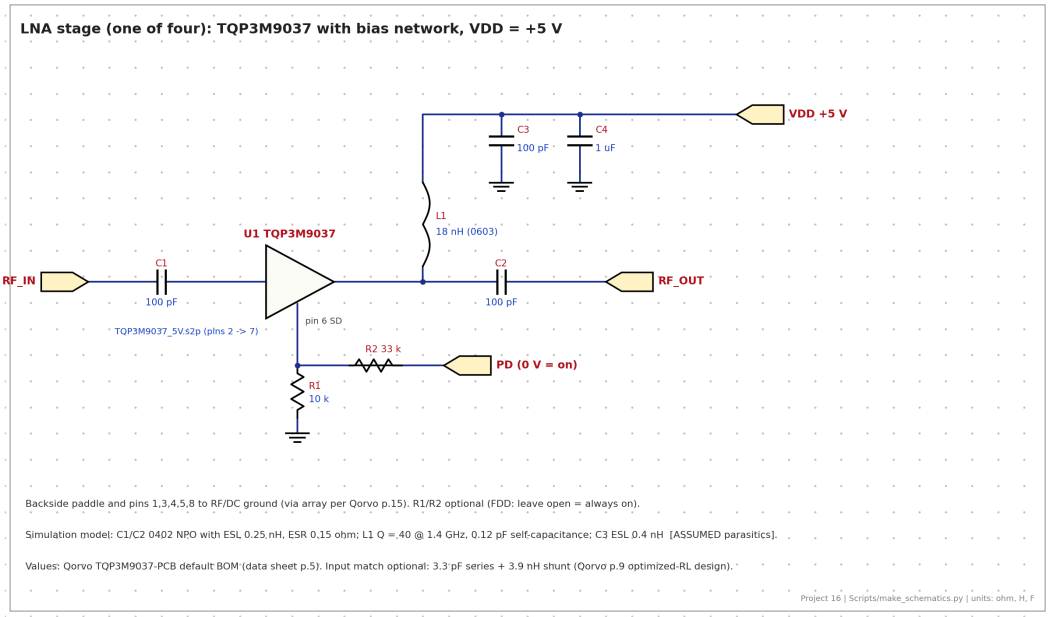


Figure 5 - L-band LNA stage with bias network (S-band stage: Qorvo 2.3-2.6 GHz reference design, 2.7 nH shunt + 2.4 pF series input match, 5.6 nH choke).

3 Performance summary

Band / MHz	Gain dB	NF dB	VSWR in	VSWR out	Ripple dB	GD ns	IIP3 dBm	OP1dB dBm	C/N0 dB-Hz
L1/E1/B1C 1575.42	38.6	0.74	2.29	1.47	0.34	38.0	-7.9	16.8	44.8
B1I 1561.10	37.7	0.74	1.49	1.35	0.51	44.4	-	-	-
G1 1602.00	37.6	0.73	2.07	1.26	0.14	41.8	-	-	-
L5/E5a 1176.45	42.0	0.59	1.63	1.42	1.14	27.3	-11.6	17.4	45.5
E5b/B2b 1207.14	41.6	0.60	2.75	2.17	0.42	24.2	-	-	-
L2 1227.60	40.1	0.62	1.96	1.58	0.41	32.0	-10.5	16.5	43.4
NavIC S 2492.028	35.0	1.17	1.24	1.15	0.08	19.9	-1.4	18.1	40.5

Table 1 - Simulated performance, 25 C typical, 50 ohm, from the antenna feed (RHCP) to the RF output. Ripple over the signal bandwidth; C/N0 for the minimum specified signal with the antenna at 290 K.

Item	L-band	S-band
Stability	unconditional; branch mu >= 1.035	unconditional; mu >= 1.053, stage mu >= 1.12
Reverse isolation	65 dB (L1), 67 dB (L5)	58 dB
Cross-pol (LHCP) rejection of hybrid	31.8 dB (L1), 34.4 dB (L5)	32.3 dB
Axial-ratio contribution of hybrid	0.45 dB (L1), 0.33 dB (L5)	0.42 dB
Output noise power (whole band)	-51.3 dBm	-59.7 dBm
3 dB bandwidth	L1 ~1553-1613 MHz, L2/L5 ~1156-1243 MHz	~62 MHz (TA1442A is wide)
Inter-band group delay	L1 - L5 = 10.7 ns (estimate)	-
Worst-case gain range (data-sheet limits)	33-45 dB, NF <= 0.98 dB	-
Supply	4 x TQP3M9037, 70 mA each at 5 V	2 x TQP3M9037, 70 mA each at 5 V
Total DC	6 LNAs: 420 mA at +5 V = 2.1 W	

Table 2 - Other parameters.

4 L-band results (simulator plots)

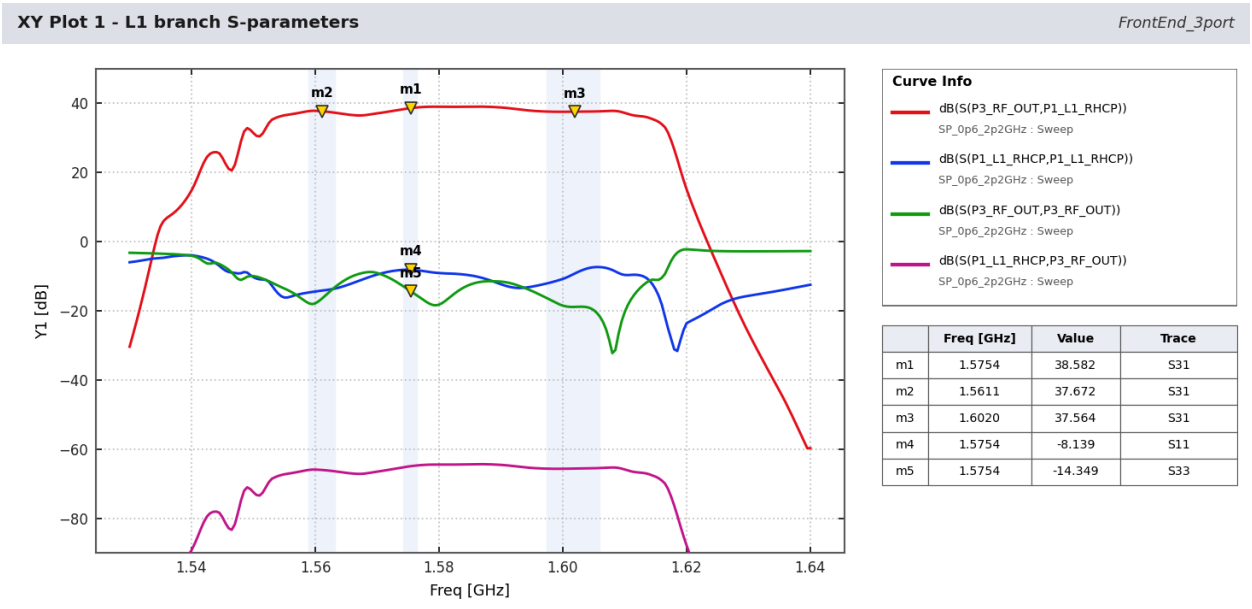
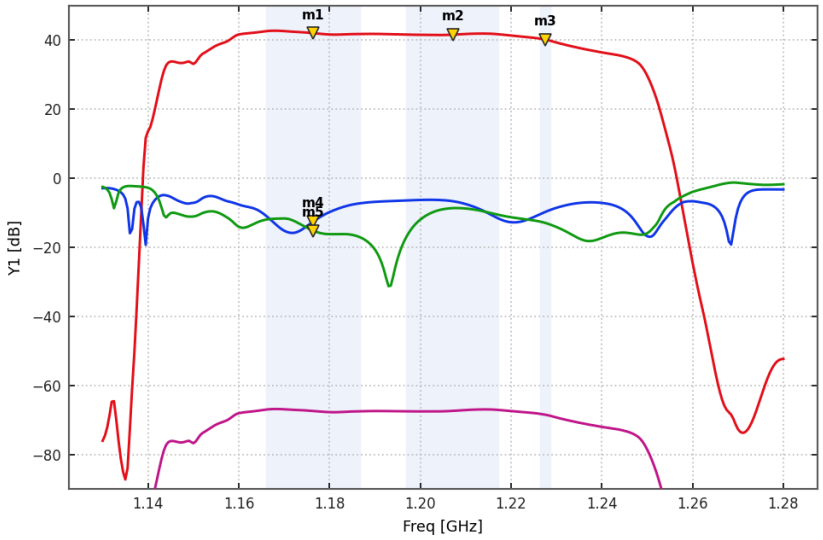


Figure - L1 branch: S31 (gain), S11, S33, S13.

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

XY Plot 2 - L2/L5 branch S-parameters *FrontEnd_3port*



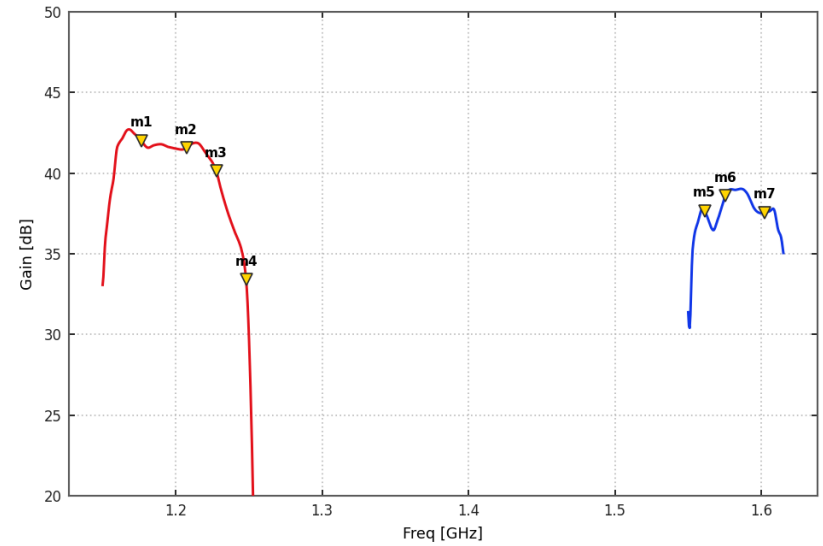
Curve Info

- dB(S(P3_RF_OUT,P2_L2L5_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P2_L2L5_RHCP,P2_L2L5_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P3_RF_OUT,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep
- dB(S(P2_L2L5_RHCP,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	41.991	S32
m2	1.2071	41.562	S32
m3	1.2276	40.115	S32
m4	1.1764	-12.498	S22
m5	1.1764	-15.216	S33

Figure - L2/L5 branch: S32, S22, S33, S23.

XY Plot 4 - Gain in the GNSS bands *FrontEnd_3port*



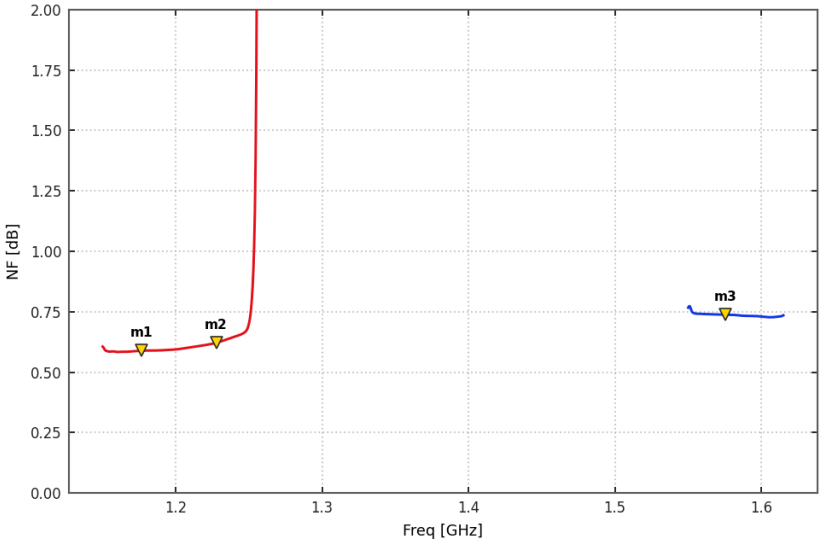
Curve Info

- dB(S(P3_RF_OUT,P2_L2L5_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P3_RF_OUT,P1_L1_RHCP)) — SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	41.991	S32
m2	1.2071	41.562	S32
m3	1.2276	40.115	S32
m4	1.2480	33.403	S32
m5	1.5611	37.672	S31
m6	1.5754	38.582	S31
m7	1.6020	37.564	S31

Figure - Gain in the GNSS bands with markers at the signal centres.

XY Plot 5 - Noise figure *FrontEnd_3port*



Curve Info

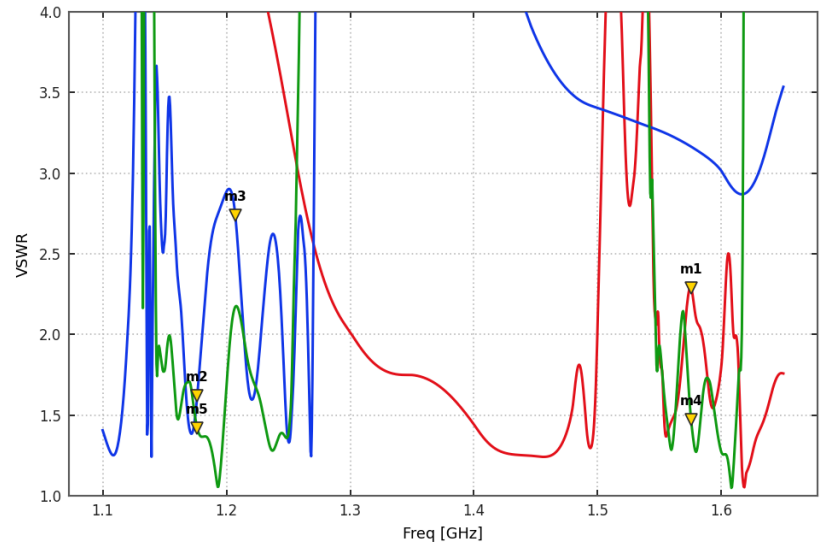
NF(P3_RF_OUT,P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

NF(P3_RF_OUT,P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	0.589	NF L2/L5
m2	1.2276	0.621	NF L2/L5
m3	1.5754	0.738	NF L1

Figure - Noise figure.

XY Plot 6 - VSWR *FrontEnd_3port*



Curve Info

VSWR(P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

VSWR(P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

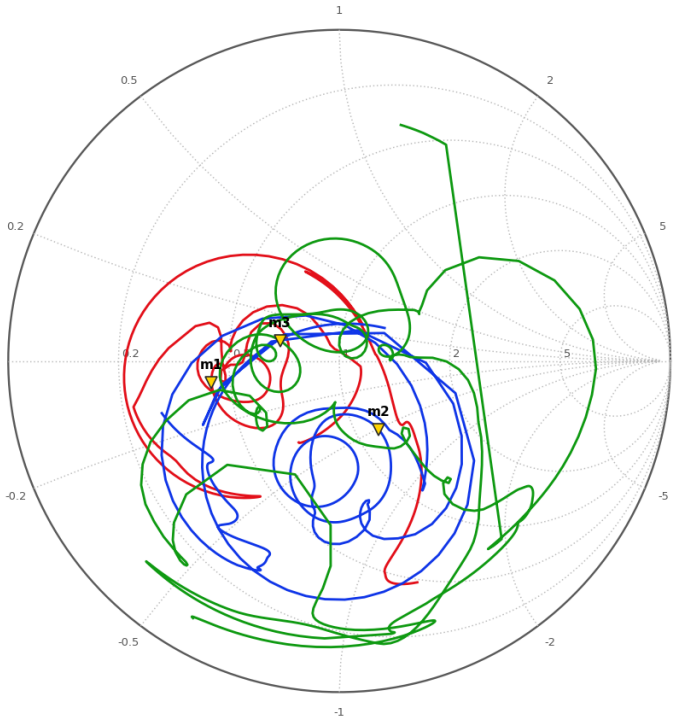
VSWR(P3_RF_OUT)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	2.288	P1
m2	1.1764	1.622	P2
m3	1.2071	2.740	P2
m4	1.5754	1.474	P3
m5	1.1764	1.420	P3

Figure - VSWR at the three ports.

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

Smith Plot 1 - Port reflections *FrontEnd_3port*



Curve Info

S(P1_L1_RHCP,P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

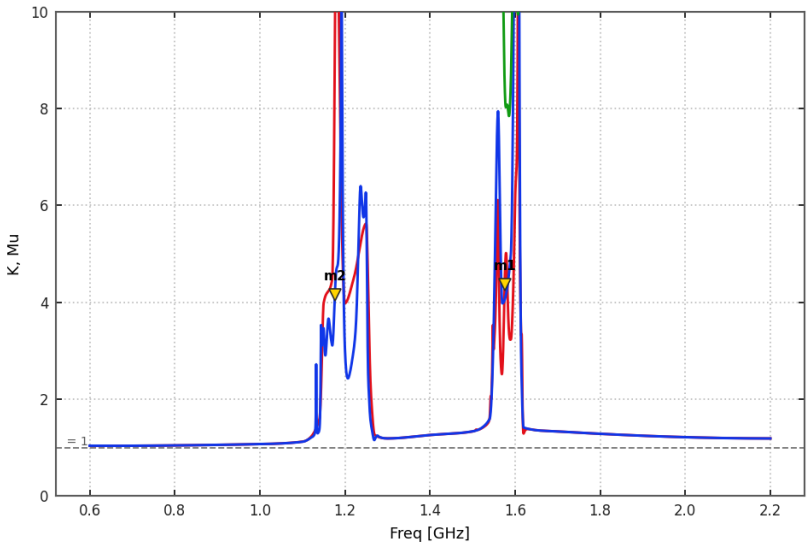
S(P2_L2L5_RHCP,P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

S(P3_RF_OUT,P3_RF_OUT)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	22.0-j3.4	S11
m2	1.1764	57.5-j25.2	S22
m3	1.5754	34.6+j4.4	S33

Figure - Port reflections on the Smith chart.

XY Plot 7 - Stability *FrontEnd_3port*



Curve Info

Mu1 (L1 branch)
SP_0p6_2p2GHz : Sweep

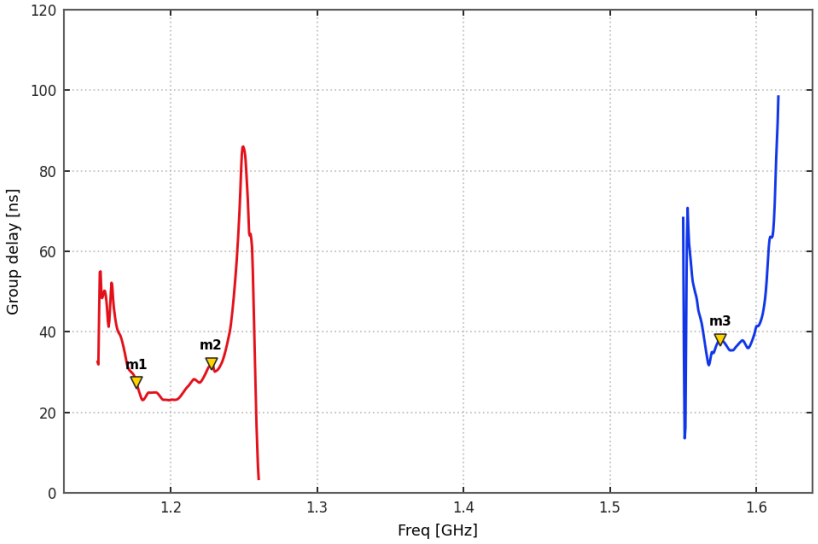
Mu1 (L2/L5 branch)
SP_0p6_2p2GHz : Sweep

StabilityFactor K (L1 branch)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	4.370	Mu L1
m2	1.1764	4.155	Mu L2/L5

Figure - Stability factors K and mu.

XY Plot 8 - Group delay *FrontEnd_3port*



Curve Info

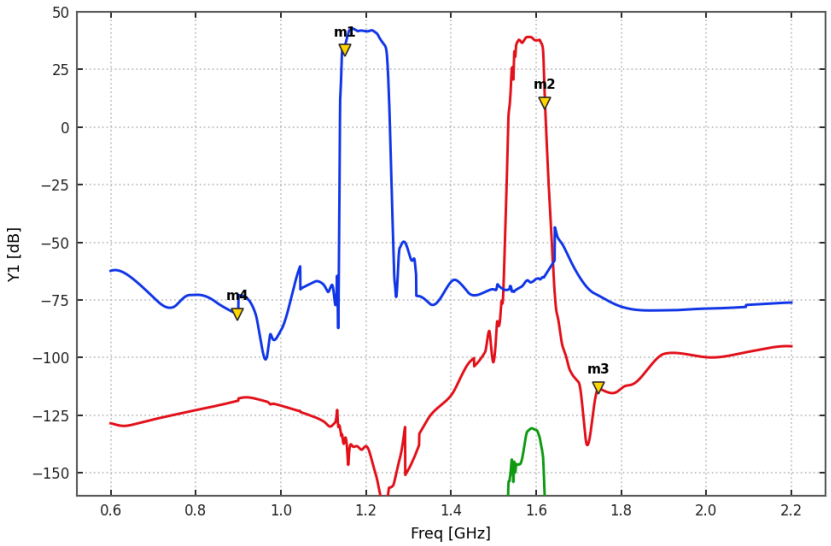
GroupDelay(P3_RF_OUT,P2_L2L5_RHCP) [ns]
SP_0p6_2p2GHz : Sweep

GroupDelay(P3_RF_OUT,P1_L1_RHCP) [ns]
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	27.353	GD L2/L5
m2	1.2276	32.102	GD L2/L5
m3	1.5754	38.021	GD L1

Figure - Group delay (estimated phase of SAW/diplexer models).

XY Plot 3 - Wide-band transmission *FrontEnd_3port*



Curve Info

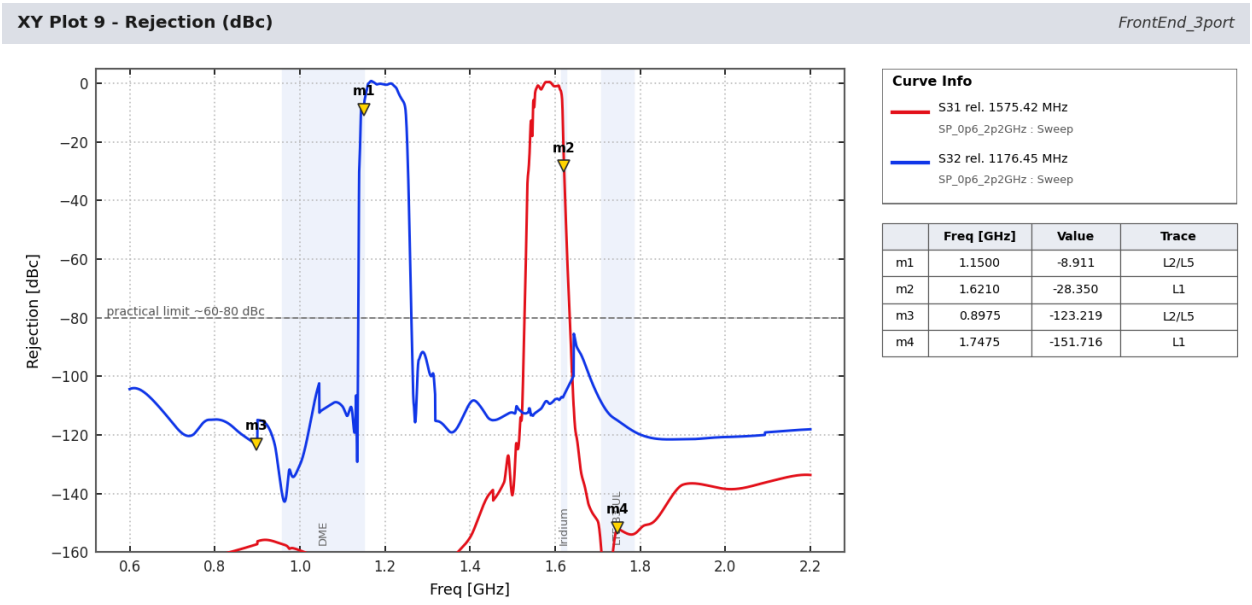
dB(S(P3_RF_OUT,P1_L1_RHCP))
SP_0p6_2p2GHz : Sweep

dB(S(P3_RF_OUT,P2_L2L5_RHCP))
SP_0p6_2p2GHz : Sweep

dB(S(P2_L2L5_RHCP,P1_L1_RHCP))
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1500	33.074	S32
m2	1.6210	10.249	S31
m3	1.7475	-113.117	S31
m4	0.8975	-81.234	S32

Figure - Wide-band transmission 0.6-2.2 GHz.



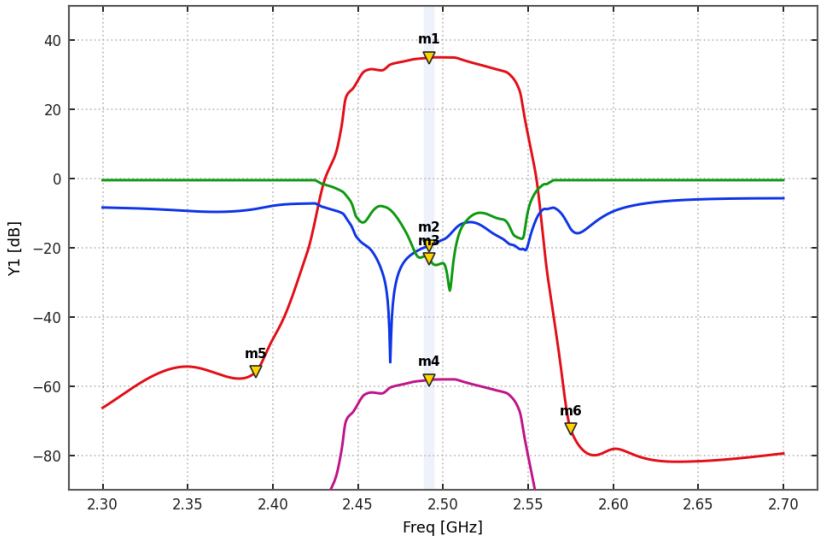
	Freq [GHz]	Value	Trace
m1	1.1500	-8.911	L2/L5
m2	1.6210	-28.350	L1
m3	0.8975	-123.219	L2/L5
m4	1.7475	-151.716	L1

Figure - Rejection in dBc relative to the band-centre gain.

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

5 NavIC S-band results (simulator plots)

XY Plot 1 - S-parameters *SBand_2port*



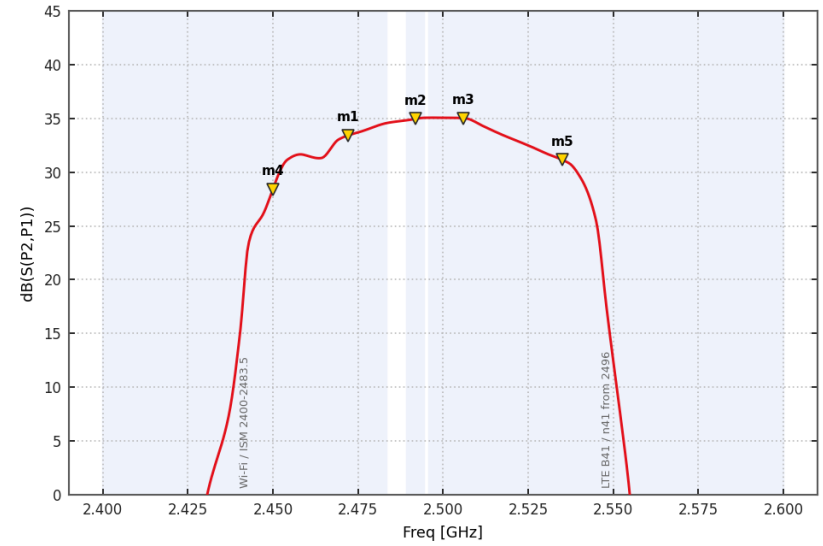
Curve Info

- dB(S(P2_S_OUT,P1_S_RHCP)) — SP_SBand : Sweep
- dB(S(P1_S_RHCP,P1_S_RHCP)) — SP_SBand : Sweep
- dB(S(P2_S_OUT,P2_S_OUT)) — SP_SBand : Sweep
- dB(S(P1_S_RHCP,P2_S_OUT)) — SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	34.954	S21
m2	2.4920	-19.393	S11
m3	2.4920	-23.280	S22
m4	2.4920	-58.184	S12
m5	2.3900	-55.874	S21
m6	2.5750	-72.437	S21

Figure - S-band branch: S21, S11, S22, S12 with markers at 2492.028 MHz and at the TA1442A specification points 2390 / 2575 MHz.

XY Plot 2 - Gain, NavIC S band *SBand_2port*



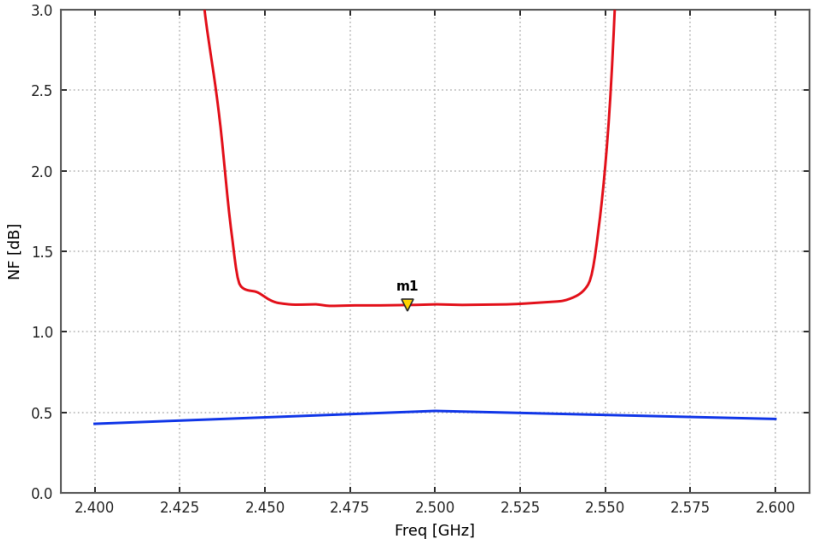
Curve Info

- dB(S(P2_S_OUT,P1_S_RHCP)) — SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4720	33.400	S21
m2	2.4920	34.954	S21
m3	2.5060	35.010	S21
m4	2.4500	28.400	S21
m5	2.5350	31.152	S21

Figure - In-band gain; markers on Wi-Fi channel 13 (2472 MHz), NavIC centre and LTE B41 (2506 MHz).

XY Plot 5 - Noise figure *SBand_2port*



Curve Info

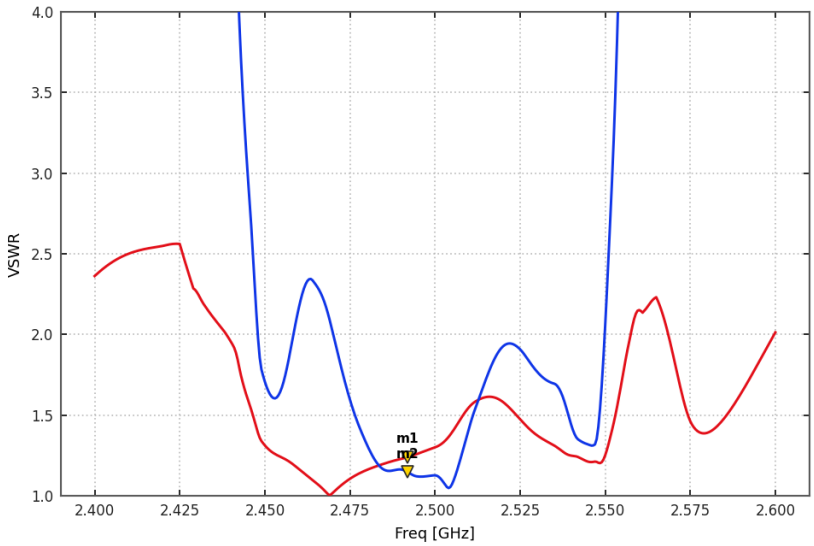
NF(P2_S_OUT,P1_S_RHCP)
SP_SBand : Sweep

NFmin TQP3M9037 (data sheet)
table, 5 V

	Freq [GHz]	Value	Trace
m1	2.4920	1.167	NF

Figure - Noise figure and LNA NFmin.

XY Plot 6 - VSWR *SBand_2port*



Curve Info

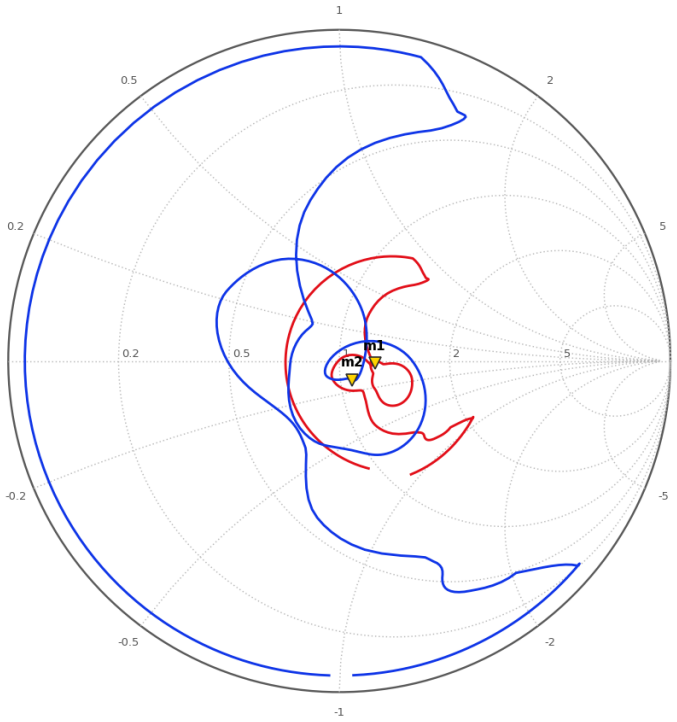
VSWR(P1_S_RHCP)
SP_SBand : Sweep

VSWR(P2_S_OUT)
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	1.240	P1
m2	2.4920	1.147	P2

Figure - VSWR.

Smith Plot 1 - Input / output reflection *SBand_2port*



Curve Info

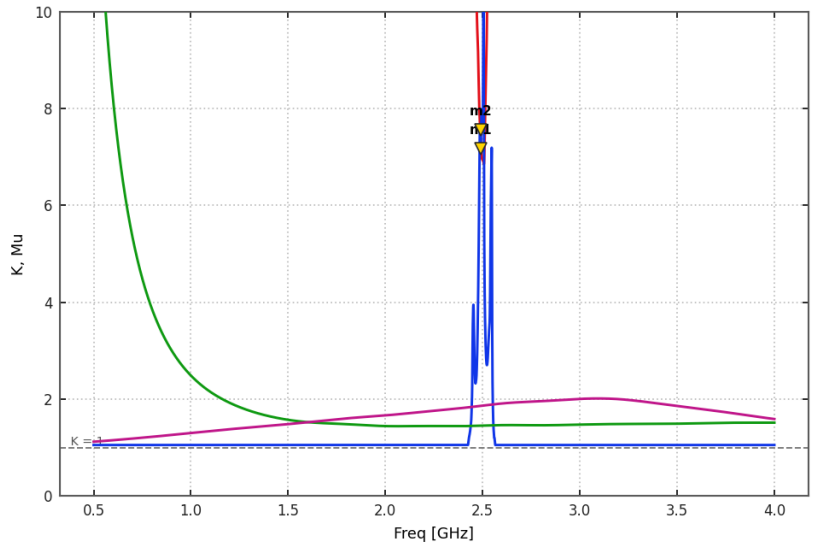
S(P1_S_RHCP,P1_S_RHCP)
SP_SBand : Sweep

S(P2_S_OUT,P2_S_OUT)
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	62.0-j0.9	S11
m2	2.4920	53.6-j6.2	S22

Figure - Input / output reflection.

XY Plot 7 - Stability *SBand_2port*



Curve Info

StabilityFactor K (system)
SP_SBand : Sweep

Mu1 (system)
SP_SBand : Sweep

K (LNA stage, ref. design)
SP_SBand : Sweep

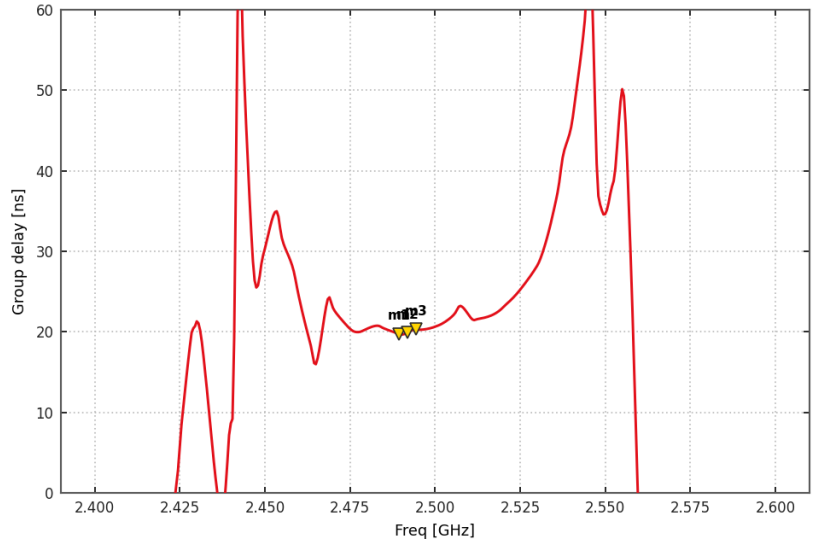
Mu1 (LNA stage)
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	7.175	K
m2	2.4920	7.560	Mu1

Figure - Stability.

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

XY Plot 8 - Group delay *SBand_2port*



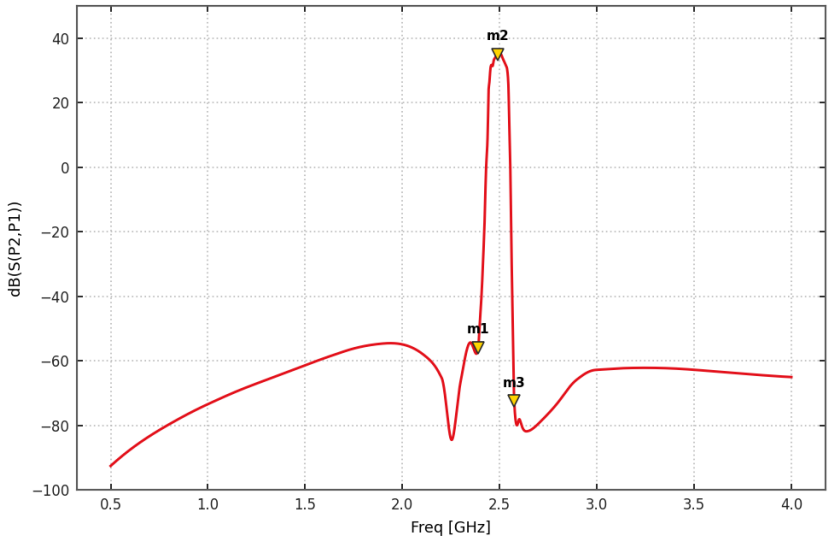
Curve Info

GroupDelay(P2_S_OUT,P1_S_RHCP) [ns]
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4895	19.737	GD
m2	2.4920	19.927	GD
m3	2.4945	20.335	GD

Figure - Group delay (estimated).

XY Plot 3 - Wide-band transmission *SBand_2port*

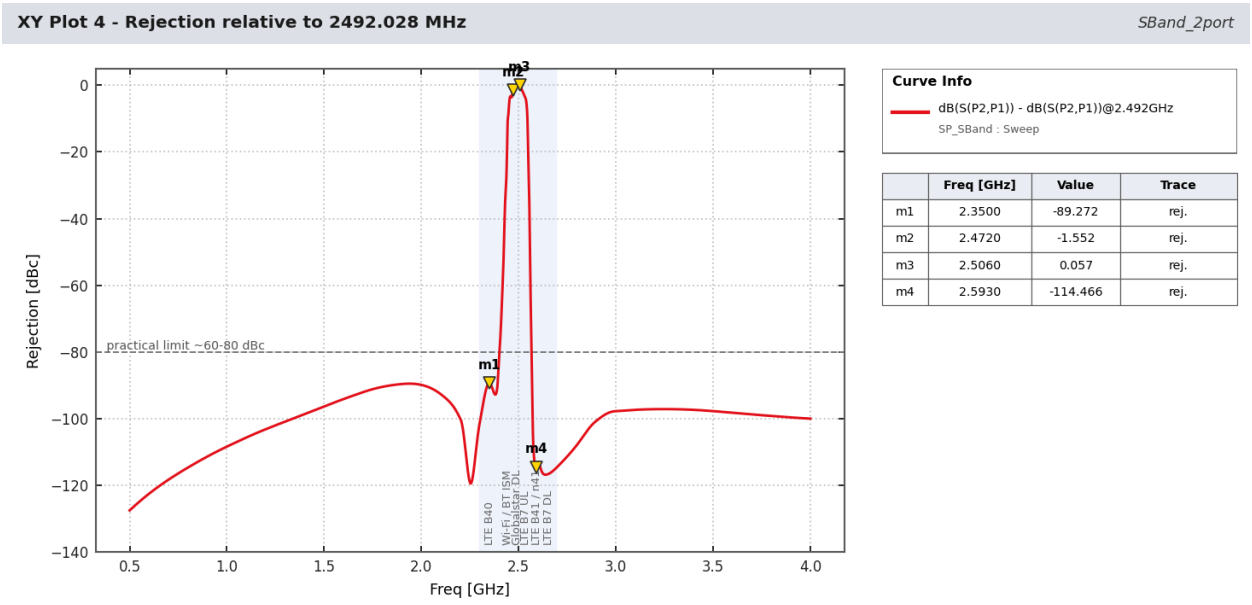


Curve Info

dB(S(P2_S_OUT,P1_S_RHCP))
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.3900	-55.874	S21
m2	2.4920	34.954	S21
m3	2.5750	-72.437	S21

Figure - Wide-band transmission 0.5-4 GHz.



	Freq [GHz]	Value	Trace
m1	2.3500	-89.272	rej.
m2	2.4720	-1.552	rej.
m3	2.5060	0.057	rej.
m4	2.5930	-114.466	rej.

Figure - Rejection relative to 2492.028 MHz.

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3 dBm	Cum IP1dB dBm	Signal dBm	Noise dBm
C2327J5003AHF (RHCP combine)	-0.43	0.43	-0.43	0.43	-	-	-132.7	-110.9
LNA1 TQP3M9037 (ref. design)	19.17	0.73	18.73	1.16	16.3	1.9	-113.6	-91.0
TA1442A #1	-1.50	1.50	17.23	1.18	16.3	1.9	-115.1	-92.5
LNA2 TQP3M9037 (ref. design)	19.17	0.73	36.40	1.19	-1.4	-15.9	-95.9	-73.3
TA1442A #2	-1.50	1.50	34.90	1.19	-1.4	-15.9	-97.4	-74.8

Table 3 - S-band cascade at 2492.028 MHz (signal = -132.3 dBm reference minimum, noise in 2.046 MHz).

6 Cascade budgets, L-band

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3 dBm	Cum IP1dB dBm	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.36	0.36	-0.36	0.36	-	-	-128.9	-110.9
LNA1 TQP3M9037 + bias	21.95	0.37	21.59	0.73	13.2	0.3	-106.9	-88.5
B7504 #1	-0.89	0.89	20.70	0.74	13.2	0.3	-107.8	-89.4
LNA2 TQP3M9037 + bias	21.95	0.37	42.65	0.74	-7.9	-20.8	-85.8	-67.5
B7504 #2	-0.89	0.89	41.76	0.74	-7.9	-20.8	-86.7	-68.4
DXP.02.A (combiner)	-3.31	3.31	38.45	0.74	-7.9	-20.8	-90.0	-71.7

Cascade - L1 C/A (1575.42 MHz, 2.046 MHz).

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3 dBm	Cum IP1dB dBm	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.19	0.19	-0.19	0.19	-	-	-128.1	-100.9
LNA1 TQP3M9037 + bias	23.61	0.39	23.41	0.58	11.2	-0.9	-104.5	-76.9
B8889 #1	-0.91	0.91	22.50	0.59	11.2	-0.9	-105.4	-77.8
LNA2 TQP3M9037 + bias	23.61	0.39	46.11	0.59	-11.6	-23.6	-81.8	-54.2
B8889 #2	-0.91	0.91	45.20	0.59	-11.6	-23.6	-82.7	-55.1
DXP.02.A (combiner)	-3.14	3.14	42.06	0.59	-11.6	-23.6	-85.8	-58.2

Cascade - L5 (1176.45 MHz, 20.46 MHz).

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3 dBm	Cum IP1dB dBm	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.22	0.22	-0.22	0.22	-	-	-130.2	-110.9
LNA1 TQP3M9037 + bias	23.39	0.40	23.17	0.61	11.4	-0.7	-106.8	-87.1
B8889 #1	-1.45	1.45	21.71	0.62	11.4	-0.7	-108.3	-88.5
LNA2 TQP3M9037 + bias	23.39	0.40	45.10	0.62	-10.5	-22.7	-84.9	-65.1
B8889 #2	-1.45	1.45	43.65	0.62	-10.5	-22.7	-86.4	-66.6
DXP.02.A (combiner)	-3.56	3.56	40.09	0.62	-10.5	-22.7	-89.9	-70.2

Cascade - L2C (1227.60 MHz, 2.046 MHz).

BEL INTERNAL - FOR INTERNAL CIRCULATION ONLY - NOT FOR EXTERNAL DISTRIBUTION

7 Interference and filtering

Interferer levels at the antenna feed are planning values for a co-located terminal (cellular / Wi-Fi uplink -10 dBm). 'Back-off' is LNA OP1dB minus the level at that LNA output; negative = LNA in compression. The TA1442A is rated +10 dBm input, the B8889 +5 dBm.

Interferer	MHz	P ant	LNA1 out	SAW1 out	LNA2 out	RF out	Back-off LNA2	SAW2 margin
LTE B40 (2300-2400)	2350.0	-10	9.1	-34.9	-15.5	-59.5	35.1	25.5
Wi-Fi ch 1	2412.0	-10	8.9	-27.2	-7.9	-44.0	27.5	17.9
Bluetooth / ISM	2441.0	-20	-1.1	-12.1	7.2	-3.8	12.4	2.8
Microwave oven	2450.0	-30	-11.2	-15.9	3.4	-1.4	16.2	6.6
Wi-Fi ch 13	2472.0	-10	8.8	6.6	25.8	23.6	-6.2	-15.8
Globalstar DL (sat.)	2490.0	-120	-101.3	-102.8	-83.6	-85.2	103.2	93.6
LTE B41 / n41 low	2506.0	-10	8.7	7.2	26.3	24.8	-6.7	-16.3
LTE B7 UL	2535.0	-10	8.6	5.4	24.5	21.3	-4.9	-14.5
LTE B41 / n41 mid	2593.0	-10	8.4	-50.1	-31.3	-89.8	50.9	41.3
LTE B7 DL	2655.0	-30	-11.8	-69.7	-50.9	-108.8	70.6	60.9
GNSS L1 (other band)	1575.4	-125	-107.5	-149.0	-131.1	-172.6	150.7	141.1
5G n78	3500.0	-20	-5.0	-51.7	-35.8	-82.6	55.4	45.8

Table 4 - S-band branch, interferer levels in dBm at each node.

Interferer	MHz	Branch	P ant	LNA1 out	LNA2 out	RF out	Back-off LNA1	Back-off LNA2
LTE B13/B14 UL	787.0	L2/L5	-10	15.0	-7.9	-88.9	7.0	29.9
GSM-900 / LTE B8 UL	897.5	L2/L5	-10	14.6	-6.6	-89.5	7.4	28.6
DME/TACAN (aviation)	1150.0	L2/L5	-30	-6.5	13.3	3.3	28.1	8.3
ARNS radar	1260.0	L2/L5	-40	-17.0	-22.8	-57.5	38.4	44.3
Inmarsat L-band DL	1545.0	L1	-80	-58.3	-43.3	-58.0	79.2	64.3
Iridium	1621.0	L1	-20	1.4	12.0	-8.3	19.4	8.8
Iridium	1621.0	L2/L5	-20	1.4	-11.7	-78.3	19.4	32.5
Inmarsat UL	1643.0	L1	-20	1.3	-7.8	-83.5	19.5	28.6
Inmarsat UL	1643.0	L2/L5	-20	1.3	-9.0	-76.6	19.5	29.8
LTE B3 UL	1747.5	L2/L5	-10	10.9	-8.1	-84.3	9.7	28.7
LTE B1 UL	1950.0	L2/L5	-10	10.2	-14.3	-92.7	9.8	34.3

Table 5 - L-band branches, cases that reach the output or load LNA 2.

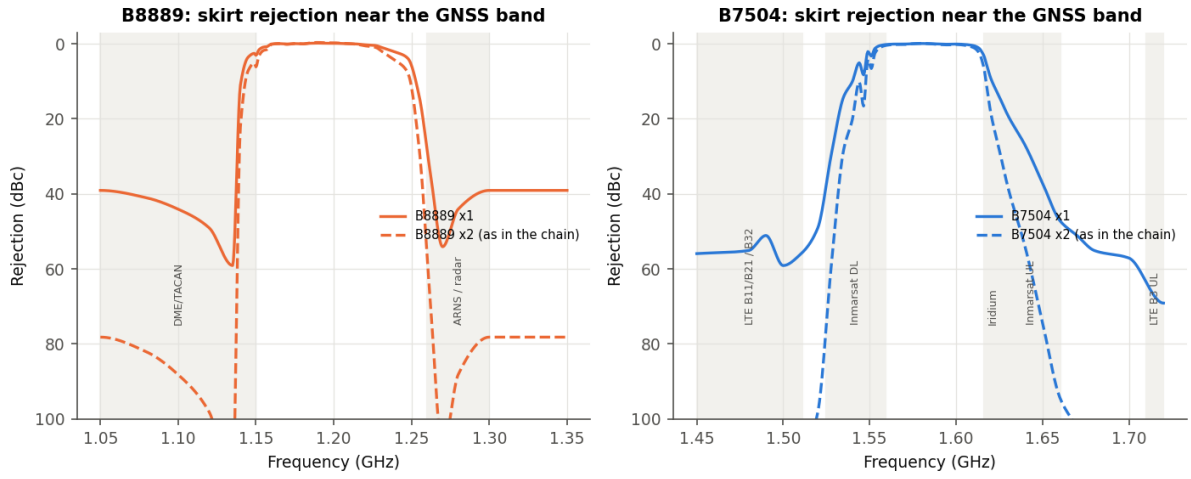


Figure - L-band SAW skirts next to the GNSS bands (one and two filters).

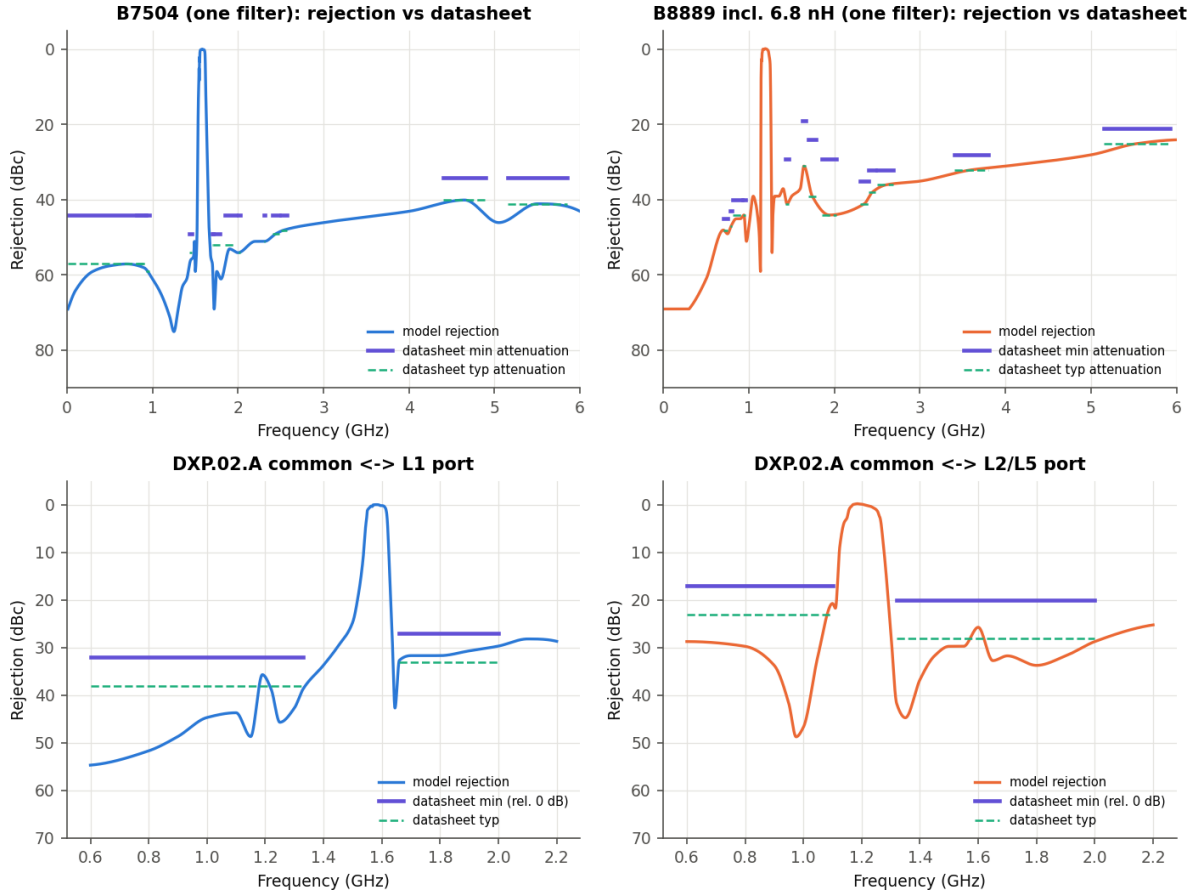


Figure - L-band filter rejection against the data-sheet limits (all met).

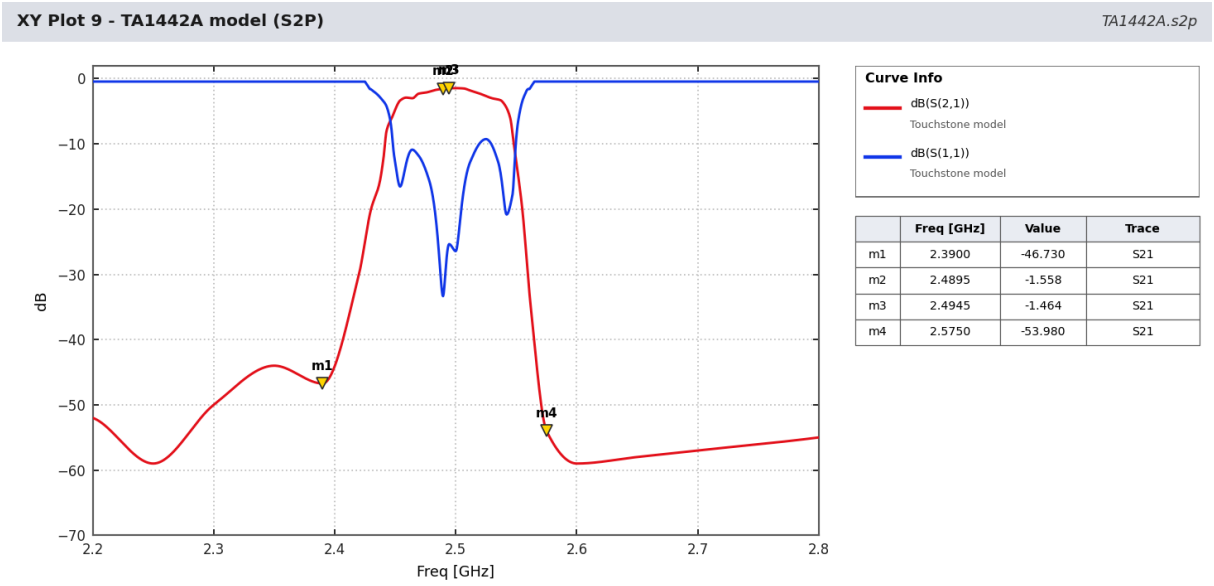


Figure - TA1442A model with the data-sheet VNA marker points (2390 / 2489.5 / 2494.5 / 2575 MHz).

8 Findings

- Noise figure is set by the hybrid and LNA 1: L1 0.74 dB, L5 0.59 dB, S 1.17 dB. The S-band hybrid (0.43 dB modelled) and the reference-design input match cost ~0.8 dB more than at L-band.
- Gain is high: 38.6 dB (L1), 42.0 dB (L5), 35.0 dB (S); worst case spread from data-sheet limits ~12 dB.
- S-band selectivity is the main risk: the TA1442A 3 dB bandwidth is ~62 MHz. A Wi-Fi channel 13 carrier at -10 dBm drives LNA 2 to +25.8 dBm (compressed, -6.2 dB back-off) and exceeds the TA1442A +10 dBm rating at SAW 2 by 15.8 dB; LTE B41 at 2506 MHz behaves the same (+26.3 dBm).

- L2/L5: DME/TACAN at 1150 MHz is passed by the B8889 lower skirt; -30 dBm gives +13.3 dBm at LNA 2 (above the B8889 +5 dBm rating).
- L1: Iridium at 1621 MHz is attenuated only ~11 dB per B7504; -20 dBm gives +12.0 dBm at LNA 2.
- Input match: L1 and E5b input VSWR 2.3-2.75 (LNA S11 reflected through the hybrid); S-band input match is good (VSWR 1.24) thanks to the reference-design input network.
- GLONASS G2 is outside the B8889 pass-band (gain falls 12 dB across G2).
- All stages unconditionally stable; three independent solvers agree to < 0.01 dB (L-band).

9 Actions / recommendations

- S-band: add a narrow pre-filter in front of LNA 1 (ceramic or cavity, <= 20 MHz, > 30 dB at 2472 and 2506 MHz), or replace the first TA1442A by a narrower NavIC S filter and move it ahead of LNA 1. Re-check NF (expect +1 dB).
- L2/L5: add a DME/L-band notch or pre-filter (> 30 dB at 960-1150 MHz). L1: steeper upper skirt for Iridium (e.g. RF360 B39162B2611P810 pre-LNA).
- Reduce gain to the receiver requirement (25-30 dB at the connector): one LNA per branch or a pad after LNA 2.
- Improve the L-band input match (Qorvo optimised-RL network or AN #18 two-step matching).
- Obtain measured S-parameter files: RF360 (B7504, B8889), Taoglas (HC125A, DXP.02.A), Tai-Saw (TA1442A), TTM (C2327J5003AHF), Qorvo (fine-step TQP3M9037 + low-band noise data). Confirm TA1442A is still in production (approval sheet dated 2011).
- Replace the ideal patch with the HFSS/measured antenna S-parameters; then axial ratio and gain/temperature can be assessed.
- Run the AEDT project (Scripts/aedt_circuit.py) on the licensed workstation to confirm in the company tool.

10 Parts and data used

Part	Maker	Function	Data used
TQP3M9037	Qorvo	LNA, 6 used	S-parameter + noise tables (5 V), eval / reference designs
HC125A	Taoglas	L-band 3 dB hybrid, 2 used	typical table
B7504	Qualcomm RF360	L1 SAW, 2 used	spec table + digitized plots
B8889	Qualcomm RF360	L2/L5 SAW, 2 used (+2 x 6.8 nH each)	spec table + digitized plots
DXP.02.A	Taoglas	L-band diplexer / combiner, 1 used	spec table + digitized plots
C2327J5003AHF	TTM (Xinger)	S-band 3 dB hybrid, 1 used	spec table + digitized plots
TA1442A	Tai-Saw	2492 MHz SAW, 2 used	spec table + digitized VNA plots
Patch elements	-	antenna (mentioned only)	ideal dual-feed RHCP source

Table 6 - Parts.

ID	Item	Assumption	How to close
A1	Topology	Two branches (patch -> HC125A -> LNA -> SAW -> LNA -> SAW) combined by the DXP.02.A, as confirmed by the user 2026-10-31. Single chain is not possible: B7504 (L1) and B8889 (L2/L5) have disjoint pass-bands.	User decision
A2	Antenna	Each patch is an ideal dual-feed RHCP source: lossless 90 deg splitter, 50 ohm feeds (PatchCP_ideal_RHCP_feed.s4p). LHCP port terminated at T0 (290 K) for noise.	Replace with measured/HFSS patch S-parameters
A3	LNA noise below 1.5 GHz	Noise table starts at 1.5 GHz. NFmin 0.38-0.40 dB, Gopt = 0.15/14.1 deg, Rn = 0.08 ohm used at 0.6-1.2 GHz (bounded by the 50-ohm eval NF of 0.40 dB at 0.9 GHz and 0.36 dB at 1.7 GHz).	Ask Qorvo for low-band noise parameters
A4	LNA bias	VDD = +5 V, 70 mA (user choice); S-parameters at device leads; eval-board bias network (100 pF DC blocks, 18 nH choke, 100 pF + 1 uF bypass).	Datasheet p.2, p.5
A5	Parasitics	100 pF 0402: ESL 0.25 nH, ESR 0.15 ohm; 18 nH 0603: Q = 40 at 1.4 GHz, 0.12 pF self-capacitance (SRF ~3.4 GHz); bypass ESL 0.4 nH.	Replace with vendor S-parameter models of the chosen parts
A6	LNA interpolation	200 MHz table interpolated with a cubic spline on dB magnitude and unwrapped phase. Linear interpolation changes in-band gain by up to 1.4 dB (heavi test) -> a real model uncertainty.	Download the Qorvo fine-step .s2p
A7	LNA large-signal	OIP3/OP1dB at L-band interpolated from eval-board values at 0.9 and 1.7 GHz (34.4-34.9 dBm, 22-20.7 dBm).	Datasheet p.6
A8	HC125A phase	Datasheet gives magnitudes and the coupled-vs-through phase difference (91.9-93.0 deg) only. Absolute through phase = -90 deg x f/1390 MHz; reflection and isolation phase offsets chosen to keep the 4-port passive (165 / 270 deg).	Measured .s4p from Taoglas

ID	Item	Assumption	How to close
A9	HC125A port mapping	'Transmission' column = pin1->pin4 (direct, angle theta), 'Coupling' column = pin1->pin3 (theta-90), from the pin table p.7. Feeds on pins 3/4, LNA on pin 1, pin 2 terminated in 50 ohm.	Confirm with Taoglas
A10	HC125A passivity	The 92 deg phase difference with 0.15-0.29 dB loss is slightly non-passive with arbitrary reflection phases; after the phase choice the matrix is scaled by <= 0.11 dB at the worst frequency.	-
A11	HC125A out of band	Values held at the 1150/1630 MHz edge outside 1150-1630 MHz. Out-of-band rejection therefore comes from the SAWs/DXP only (conservative).	-
A12	SAW phase	No measured phase in the datasheets. S21 phase = minimum-phase (Hilbert / real-cepstrum) reconstruction of the digitized magnitude; S11/S22 phase = S21 phase + 90 deg (lossless symmetric reciprocal condition). Group delay is therefore an estimate; ripple agrees with the brief (B7504 GD ripple 4 ns typ).	Use RF360 measured .s2p (available on rffe.qualcomm.com)
A13	SAW reflection out of band	|S11| = 0.95 sqrt(1-|S21|^2) outside the plotted VSWR range (reflective SAW stop-band), blended over 4 MHz.	-
A14	B8889 matching	The B8889 model includes the required 6.8 nH shunt inductors (datasheet curves are measured with them).	Datasheet p.6
A15	DXP model	Digitized curves include the Taoglas eval-board matching (L2 4.3 nH, C1 1.5 pF, L1 4.7 nH, L3 3.3 nH, C2 1 pF). Reflection phase = dominant transmission phase + 90 deg.	Measured .s3p from Taoglas
A16	Digitizing accuracy	Pass-band plots +/-0.3 dB, wide-band attenuation plots +/-2 dB, VSWR +/-0.1.	-
A17	PCB	Interconnects ideal (zero length). 50 ohm lines on the real board add ~0.02-0.05 dB/cm loss and phase.	Add MSL models once the layout exists
A18	Interferer levels	Interferer powers at the antenna feed (-10 dBm cellular UL, -20 dBm Iridium/Inmarsat UL, -30 dBm DME) are example values for a co-located handset/terminal.	Set from the product's EMC environment
A19	Signal levels	Minimum received power (0 dBic antenna): GPS L1 C/A -128.5 dBm, L5 -127.9 dBm, L2C -130 dBm, GLONASS G1 -131 dBm (signal ICDs).	-
A21	S-band topology	S-band patch (ideal dual feed) -> C2327J5003AHF -> TQP3M9037 -> TA1442A -> TQP3M9037 -> TA1442A -> separate S-band output (no diplexer). Antenna only mentioned, modelled as ideal RHCP source.	User request 2026-10-31
A22	S-band LNA match	Qorvo 2300-2600 MHz reference design (2.7 nH shunt, 2.4 pF series, 5.6 nH choke); inductor Q 30-50 @2.5 GHz, ESL/ESR [ASSUMED].	Datasheet p.10
A23	C2327J5003AHF model	Magnitudes from excess IL, amplitude balance, isolation and input return loss (same RL assumed at all ports); coupled phase = direct + 90 deg - phase balance; reflection/isolation phases chosen for passivity over 2-3 GHz -> excess IL 0.43 dB vs 0.3 typ / 0.4 max (conservative).	Measured .s4p from TTM
A24	TA1442A model	Digitized VNA plots p.3-4 with the printed marker values used exactly; S11 = S22 shape (markers differ by < 0.01); min-phase phase.	Measured .s2p from Tai-Saw
A25	NavIC S signal	2492.028 MHz, SPS BW 2.046 MHz, RS BW 16.5 MHz, minimum received power -132.3 dBm (0 dBic) used as reference values.	Verify against the IRNSS SPS ICD
A20	Temperature	All data at 25 C typical. No temperature, part-to-part or aging spread is modelled.	-

Table 7 - Assumptions.

ID	Where	Finding
I1	B7504 vs product brief	Datasheet p.7 lists 'Group delay typ 4.0 / max 15 ns' (GLONASS band); the product brief lists the same numbers as 'group delay ripple'. The model gives ~13 ns absolute delay and ~4-10 ns ripple, so the datasheet label should read 'ripple'.
I2	DXP.02.A front page	Lists 'L2 1227.625 MHz'; GPS L2 is 1227.60 MHz.
I3	DXP.02.A isolation row	'Isolation (1574.22~1576.62 dB)' - unit should be MHz.
I4	DXP.02.A pass-band claim	Text p.11: L2 pin covers '1164-1300 MHz'. Measured plot p.8: insertion loss >6 dB above ~1263 MHz, so GLONASS G2 upper half, B3 (1268.5) and E6 (1278.75) are not usable.
I5	DXP.02.A return loss	Table p.4 band-2 RL typ -8 dB (max -6) vs. the plot p.6 showing 12-24 dB in the L2/L5 band; the plot is used.
I6	DXP.02.A pin naming	Pin 5 is called 'GPS L2 Output' in the pin table but 'GPS L2/L5 OUT' on the symbol.
I7	DXP.02.A RL sign	Return loss given as negative numbers with 'Max -6.0' - ambiguous min/max sense.
I8	HC125A phase balance	Spec says 'Phase balance 90 degrees' with no tolerance; the typical table shows 91.9-93.0 deg (up to 3 deg off), which costs ~0.01 dB of axial-ratio/combining loss.
I9	HC125A pin table vs table columns	Typical table uses 'Coupling/Transmission' and S11..S44 without stating which pins; the pin table uses 'Pin 1..4'. Mapping assumed (A9).
I10	TQP3M9037 noise table	NFmin 0.40 dB at 1.5 GHz but 0.29 dB at 1.7 GHz (non-monotonic); Gopt angle jumps 85.8 -> 42.1 -> 157.4 deg between 2.4/2.5/2.6 GHz with |Gopt| ~0.02-0.08 (measurement scatter).
I11	TQP3M9037 switching speed	Table p.2 note: switching speed with C1 = 10 pF; eval board default C1 = 100 pF (switching table p.5 shows both).
I12	TQP3M9037 stability	'Unconditionally stable' claim holds (K >= 1.01, mu >= 1.06 from the table) but the margin is only ~1 % at 50 MHz.
I13	B8889 spec typo	Data sheet p.7 'Insertion attenuation' first row has no unit; 'linsertion loss' typo on B7504 p.7.
I14	B8889 vs brief	Brief gives B8889 pass-band 1166-1228 MHz (62 MHz); the datasheet spec bands end at 1228.62 MHz. GLONASS G2 (1246-1256) is outside the B8889 pass-band (>4 dB at 1245 MHz) although the HC125A table and the DXP claim to cover it.

ID	Where	Finding
I15	SDST vs. parts used	The automotive SAW selection table (Feb 2020) does not list B7504 or B8889; equivalent automotive parts are e.g. B39162B2611P810 (L1, 0.9 dB, pre-LNA) and B39122B2620P810 (E5b/L2).
I17	DXP.02.A insertion loss	Table p.4: IL typ 3.8 dB (L1) / 4.2 dB (L2/L5); the measured plots p.7-8 show 3.3 dB / 3.1-3.2 dB at the band centres and reach 3.8 / 4.2 dB only at the band edges. The model follows the plots.
I18	TA1442A bandwidth	Cover page 'BW = 5 MHz' is the specified band 2489.5-2494.5 MHz; the measured 3 dB bandwidth is ~62 MHz (about 2453-2537 MHz) and the filter passes Wi-Fi channel 13 and LTE B41/B7 almost unattenuated.
I19	TA1442A VSWR	Spec VSWR typ 1.2 vs VNA markers 1.04-1.12 in the spec band (sample better than typ).
I20	C2327J5003AHF note	Spec table footnote: 'values are for the case with shunt capacitor across differential lines' - not explained for a 4-port hybrid; no value given.
I21	C2327J5003AHF phase balance	Spec typ 4 deg / max 8 deg vs plot ~2.5-3.2 deg over 2.4-2.6 GHz; broadband plot shows >= 20 deg above 4.4 GHz (part not usable there).
I22	Two datasheet ages	TA1442A approval sheet is from 2011 (rev. 1): check the part is still in production before design-in.
I16	AN #18 scope	AN #18 covers 315/434/868 MHz RKE receivers (TDA520x, ATA572x). Only its matching method applies here; its component values do not.

Table 8 - Data-sheet inconsistencies.

Simulation files: Touchstone/ (component models and simulated results), Scripts/ (Python models, PyAEDT script with designs FrontEnd_3port and SBand_2port), circuitRF/ (netlist), Extracted_Data/ (parameter tables, results).